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Predictive Analytics in Human Resources Management: Evaluating AIHR's Role in Talent Retention

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Abstract

This study explores the role of artificial intelligence (AI) in human resource management (HRM), with a focus on recruitment, employee retention, and performance optimization. Through a PRISMA-based systematic literature review, the paper examines many machine learning algorithms including XGBoost, SVM, random forest, and linear regression in decision-making related to employee-attrition prediction and talent management. The findings suggest that these technologies can automate HR processes, reduce bias, and personalize employee experiences. However, the implementation of AI in HRM also presents challenges, including data privacy concerns, algorithmic bias, and organizational resistance. To address these obstacles, the study highlights the importance of adopting ethical AI frameworks, ensuring transparency in decision-making, and developing effective integration strategies. Future research should focus on improving explainability, minimizing algorithmic bias, and promoting fairness in AI-driven HR practices.

Keywords: artificial intelligence; AIHR; machine learning; human resource management; employee attrition; predictive analytics; talent retention

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1. Introduction

Human resource management (HRM) plays a strategic role in enhancing organizational effectiveness, innovation, and competitiveness. Among its core functions, recruitment, employee retention, and performance optimization are critical for maintaining a productive and motivated workforce. Organizations that fail to manage these elements effectively often face high turnover costs, reduced productivity, and difficulties in sustaining long-term growth [1,2].

Employee retention continues to be a pressing challenge across industries. High attrition rates negatively impact organizational performance by increasing the costs associated with hiring, training, and lost knowledge [3,4]. Research indicates that factors such as job satisfaction, work–life balance, career advancement opportunities, and alignment with organizational values influence an employee's decision to stay or leave [5]. HR departments require tools that not only identify high-risk individuals but also support targeted interventions to prevent turnover.

Performance optimization involves the ongoing evaluation, feedback, and development of employees to ensure they meet organizational goals. Effective performance management strategies are essential for employee motivation, productivity, and long-term career growth [6]. However, performance metrics are often subjective, and manual assess-

ments may overlook important behavioral patterns or learning needs, thus limiting the effectiveness of traditional approaches [7].

Artificial intelligence (AI) and machine learning (ML) have emerged as transformative tools in HRM, offering predictive analytics and automation for critical HR tasks. AI refers to the broader theory and development of computer systems that perform tasks typically requiring human intelligence, such as decision-making or pattern recognition. ML, a subset of AI, enables systems to learn from data and generate inferences autonomously without explicit programming [8]. These technologies are now being adopted to enhance recruitment processes, forecast employee attrition, and personalize development plans based on real-time data.

This study differs from previous research in several key ways. While earlier studies have typically focused on applying individual machine learning algorithms to predict employee turnover, this investigation adopts a comparative, model-based approach, analyzing the predictive performance of a range of algorithms—including logistic regression, random forest, ExtraTreesClassifier, LightGBM, and XGBoost—particularly in the context of unbalanced HR datasets. Furthermore, the existing literature is extended by integrating concepts from organizational psychology, such as the three-component model of organizational commitment (affective, continuance, and normative), and by assessing their impact on the predictive accuracy of AI models. Lastly, the study places special emphasis on interpretability and the practical relevance of results for HR strategy development.

The goal of this study is to identify research gaps in the use of AI for human resource applications, where researchers in the field can contribute new insights, and to highlight the limitations of existing techniques in order to enhance the robustness of future solutions.

The contribution of our paper in this field is twofold:

- Firstly, based on a PRISMA adapted methodology we selected the most impressive works that emphasize machine learning-based techniques applied to human resource management.
- Secondly, we identify and compare AI algorithms used in HR applications to underline the most robust ones for this topic and to discover in this way the gaps where new hybrid solutions can be further developed for a robust AIHR approach.

Despite the increasing use of machine learning in HRM, existing literature often lacks comparative evaluations of models applied in real-world HR contexts, and there is limited discussion of how these technologies align with HRM theories [3–9]. This paper addresses this gap by systematically reviewing recent studies and evaluating the strengths, limitations, and applicability of different AI approaches in HR tasks such as attrition prediction and recruitment. This study builds upon prior research that has demonstrated the effectiveness of AI models such as random forest, XGBoost, and neural networks in predicting employee attrition and optimizing HR processes. However, unlike previous work, which often focuses on single models or synthetic datasets, this paper offers a comparative review of multiple models evaluated across various HR scenarios, thus contributing to a more holistic understanding of AI's potential in human capital management.

Although this study does not aim to introduce novel algorithms or fundamentally new methodologies, its primary objective is to provide a structured and interdisciplinary comparative analysis of widely adopted artificial intelligence techniques—specifically XGBoost, random forest, and logistic regression—as applied to employee attrition prediction in human resource management (HRM).

The contribution lies in synthesizing recent empirical findings across real-world datasets and evaluating model performance using robust metrics such as AUC and accuracy, particularly in the context of high-dimensional and imbalanced HR data. Furthermore, the integration of organizational psychology frameworks—such as organizational commitment

theory and expectancy theory—enhances the interpretability of the selected models by linking algorithmic outcomes to psychological and motivational constructs. This interdisciplinary alignment is rarely addressed in purely technical literature and serves as a bridge between data-driven insights and organizational behavior.

Additionally, the paper emphasizes practical aspects of implementation, including data preprocessing strategies, feature selection through Pearson correlation, class balancing using techniques such as ADASYN and undersampling, and the tuning of model hyperparameters to optimize predictive performance. These methodological considerations help adapt existing models to the specific challenges found in HR contexts, where data imbalance and interpretability often present significant constraints.

While the approach may not represent a methodological breakthrough, it offers a valuable practice-oriented contribution by aligning technical rigor with domain-specific applicability. This makes the findings more actionable for both HR practitioners and academic researchers seeking to leverage AI in talent management and retention.

2. Materials and Methods

The most suitable articles were selected using the PRISMA methodology [10], which provides a structured framework for conducting systematic literature reviews. We applied the PRISMA technique, as it is the most suitable for any systematic review of the specialized literature, being able to retain only the most relevant articles from large databases.

We selected IEEE Xplore and Google Scholar because they are two of the most comprehensive and accessible academic databases in the fields of computer science and artificial intelligence. IEEE Xplore provides peer-reviewed, high-quality conference and journal papers on technical subjects, while Google Scholar aggregates a broader range of research sources, allowing us to capture interdisciplinary studies relevant to both AI and human resource management. The following keyword combinations were used as exact search terms: "Machine learning AND HR recruitment", "Artificial intelligence algorithm for HR", "Artificial intelligence AND recruiting process", and "Machine learning AND management human resources".

These terms were chosen to capture research published at the intersection of AI techniques and human resource applications, particularly in areas such as recruitment, employee retention, and performance optimization.

The search was limited to peer-reviewed journal and conference publications written in English and published between January 2019 and March 2024. A total of 1200 articles were initially retrieved. The exclusion criteria included: (1) duplicate articles, (2) papers that did not match our keywords based on title and abstract, and (3) studies that lacked technical relevance or did not provide empirical results. After removing 150 duplicate articles, the remaining set consisted of 1050 articles. Following a review of these studies based on their titles and abstracts, 950 articles were excluded. From the remaining 100 articles, 71 were excluded after in-depth reading, based on exclusion criteria focusing on articles retained for analysis and research. The resulting 29 articles met the research criteria for in-depth analysis.

The goal of this research is to address the research questions by identifying all relevant research findings from prior studies. The research questions are organized into three questions:

Q1. What AI-driven approaches have been implemented in human resource solutions?

Q2. What models have been designed to enhance database analysis for HR solutions?

Q3. How can the efficiency of HR solutions be optimized?

Thus, the most suitable articles were selected using the PRISMA technique (Preferred Reporting Items for Systematic Reviews and Meta-Analyses), as shown in Figure 1. By

applying the PRISMA technique, which is appropriate for any systematic review of the literature, only the most relevant articles from large databases were retained. In the final step, as shown in Figure 1, the final set of articles was not solely the result of automatic selection based on keyword combinations but also represented responses to the research topic:

1. Articles that use the most advanced analysis techniques.
2. Articles written in English.
3. Articles published in the last four years (2019–2024).

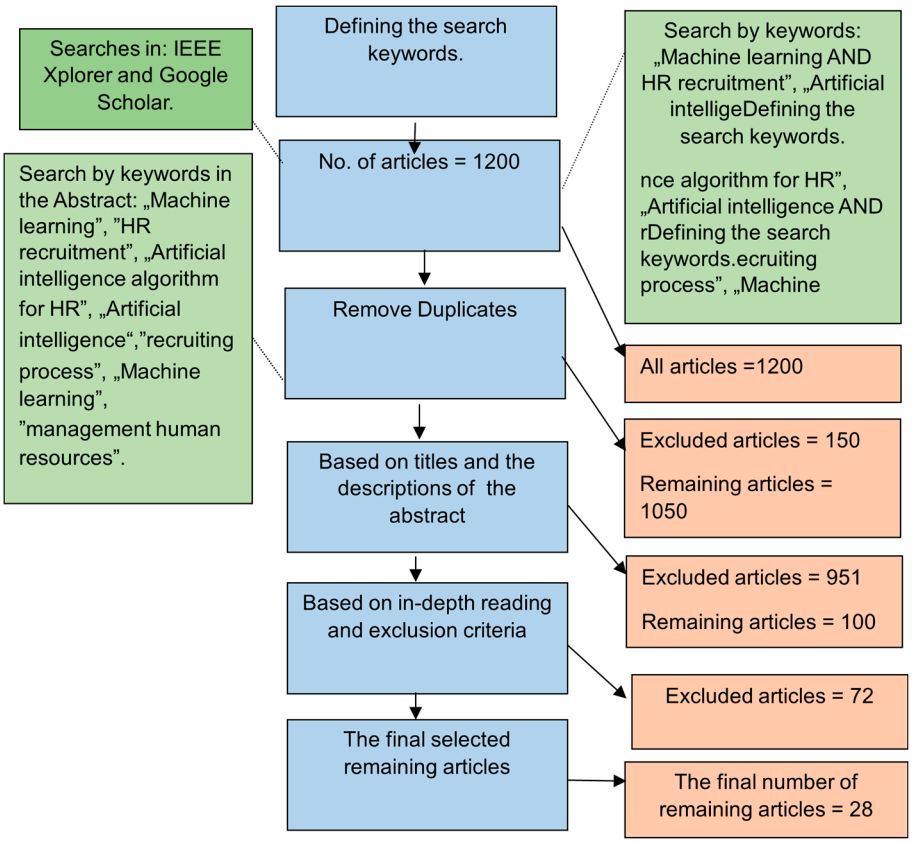

Figure 1. Article filtering methodology.

Subsequently, the following exclusion criteria were applied:

1. Duplicate references from different electronic archives searched.
2. Articles that did not address any of the research questions.
3. Articles that did not align with the study objectives.
4. Articles whose abstracts did not match the keywords from the search criteria.

After conducting a systematic search in scientific databases, we identified the most relevant studies that addressed the application of AI and machine learning techniques in human resource management. The selected articles focus on key HR processes such as recruitment, employee retention, and performance optimization. Specifically, they analyze how AI algorithms are used to predict employee attrition, improve candidate selection, and support performance evaluation and engagement tracking. These studies form the empirical foundation for the comparative analysis presented in this review.

We carefully analyzed the selected articles and evaluated the extracted findings, synthesizing the existing research to identify the most effective techniques and highlight potential areas for future exploration in this field.

3. Context of Management Theory Framework

To position the current research within the broader context of management theory, we draw on several foundational concepts from human resource management. Organizational commitment theory [3] explains employee retention through three components: affective, continuance, and normative commitment—each influenced by factors such as job satisfaction, career development, and alignment with organizational values. Recent studies have highlighted that organizational commitment does not manifest uniformly across all organizational contexts, which directly impacts how AI models for employee retention should be designed and trained. The research in [11] indicates that employees belonging to different generational groups demonstrate distinct organizational commitment profiles, with younger age groups generally exhibiting lower levels of normative commitment compared to their older counterparts. Furthermore, studies comparing public and private sectors [12] show that affective commitment is more prevalent in private organizations, while continuance commitment dominates in institutions characterized by high job stability. Evidence from [13] suggests that most current AI systems fail to account for these distinctions, often treating commitment as a uniform factor, which can ultimately reduce prediction accuracy. From a forward-looking perspective, a 2024 study [14] explores how generative AI—when trusted—can enhance perceived organizational support and strengthen affective commitment. Taken together, these findings suggest that integrating differentiated commitment constructs into AI models may enhance retention prediction performance by aligning model logic with the psychological diversity of employees. Building on this, AI models can also identify behavioral patterns that may indicate low commitment, such as frequent absences or reduced engagement. The expectancy theory [15] suggests that employees are motivated when they believe their effort leads to performance, performance leads to rewards, and those rewards are personally valued. Machine learning algorithms can detect signals of diminished motivation by analyzing variables such as delayed promotions or lack of recognition. Equity theory [9] emphasizes employees' perceptions of fairness in the ratio between their inputs (effort) and outcomes (rewards), relative to others. Predictive models can highlight discrepancies in salary, workload, or recognition that may lead to perceived inequity and dissatisfaction. These theories provide a conceptual foundation for understanding and interpreting the predictive outputs of AI systems in HRM. Aligning machine learning insights with well-established motivational theories enhances both the analytical rigor and ethical relevance of AI applications in human capital management.

In order to strengthen the empirical relevance of the theoretical framework discussed above, we mapped the psychological and organizational constructs derived from organizational commitment theory, expectancy theory, and equity theory to actual variables in the IBM Employee Attrition dataset [16]. This mapping enabled a more systematic integration of management theory into AI model design and interpretation, as detailed below: affective commitment is reflected in features such as JobSatisfaction, WorkLifeBalance, and EnvironmentSatisfaction, which capture employees' emotional attachment and workplace experience. Continuance commitment is approximated through YearsAtCompany, YearsInCurrentRole, and TotalWorkingYears, which represent tenure-based investment and switching costs. Normative commitment is indirectly indicated by features such as OverTime, TrainingTimesLastYear, and PerformanceRating, which reflect employee loyalty and perceived duty.

Expectancy theory's components—effort, performance, and reward—are operationalized through variables like OverTime (effort), PerformanceRating (performance), and MonthlyIncome or StockOptionLevel (reward). Similarly, Equity Theory is represented by features such as EnvironmentSatisfaction, JobInvolvement, and HourlyRate, which help assess the balance between perceived input and organizational outcomes.

These theory-driven features were used to evaluate model performance and interpret AI outputs. For instance, models like XGBoost, which handle non-linear relationships, better capture the complex interactions between commitment or reward perceptions and turnover likelihood. This theoretical grounding adds depth to our feature selection process and supports the practical relevance and interpretability of AI predictions in real HR contexts.

From a modeling perspective, variables such as *JobSatisfaction*, *EnvironmentSatisfaction*, and *WorkLifeBalance* can serve as proxies for affective commitment, while *YearsAtCompany* and *YearsWithCurrManager* may reflect continuance commitment. Features like *OverTime* or *PerformanceRating* relate to perceived effort and reward dynamics, aligned with expectancy principles. These theoretical underpinnings support more informed feature selection and interpretation of model outcomes.

Additionally, the applicability of these theories varies by organizational context. As shown by Murire (2024) [17], leadership style, organizational values, and communication culture shape how employees perceive fairness, expectations, and commitment. For example, mission-driven organizations may place greater weight on intrinsic motivators (e.g., purpose, learning), reinforcing affective and normative commitment. In contrast, performance-driven organizations may rely more on outcome-based motivators, making equity and expectancy theories more relevant for predictive modeling. Embedding this theoretical awareness into AI-based systems can enhance both the ethical grounding and practical impact of predictive analytics in HRM.

4. Introducing Artificial Intelligence in Human Resources Analytics

Artificial intelligence is playing an increasingly strategic role in modern recruitment processes. As companies compete to attract top talent, AI-powered solutions such as resume screening tools, chatbots, and algorithmic matching systems help optimize candidate selection and enhance the applicant experience. Organizations that successfully integrate AI technologies into recruitment can reduce time-to-hire, eliminate bias in early screening stages, and offer personalized communication at scale [18]. As a result, companies that align their recruitment strategies with digital expectations are more likely to secure high-quality candidates and maintain a competitive advantage in the evolving job market.

While AI offers considerable efficiency in recruitment—automating tasks such as resume screening, scheduling, and candidate ranking—research shows that candidates still prefer human interaction at key stages of the hiring process. AI-based recruitment strategies must therefore be complemented by strong employer branding and personalized communication. Addressing candidate expectations through hybrid approaches that blend automation with human touchpoints can significantly improve engagement and enhance the candidate experience [19,20].

AI technologies are increasingly leveraged in recruitment processes to streamline tasks and improve efficiency. For example, chatbots and virtual assistants powered by natural language processing (NLP) can automate candidate interactions, answer frequently asked questions, and convert voice inputs into text, reducing manual workload for recruiters. Robotic process automation (RPA) further supports talent acquisition by gathering resumes from online platforms and structuring candidate information for screening [21]. Once collected, AI algorithms—particularly those using string-matching or semantic similarity techniques—assist in identifying candidates whose profiles best align with specific job requirements [22,23]. These tools not only accelerate the recruitment process but also contribute to improved candidate–job fit and reduced hiring bias.

The ability to identify, analyze, and protect employee-related data is essential for leveraging AI to improve retention strategies [24]. Optimal states of relevant HR data,

can be identified through the analysis of machine learning models trained on historical data and subjected to explanation methods. These methods provide recommendations on actions to be taken to improve employee retention [25]. This data-driven approach supports proactive retention strategies tailored to individual and organizational contexts.

Human resource departments increasingly use AI-powered virtual assistants and software bots to support the onboarding process. These tools interact with new employees by providing timely information about company policies, benefits, and procedures, and assist in answering common questions. Automating onboarding in this way enhances the employee experience from the outset, reducing confusion and improving engagement—two factors closely linked to early-stage retention.

A key strategic consideration for HR departments is how to effectively integrate artificial intelligence into workforce management to improve employee experience—a critical driver of retention and engagement. Machine learning enables organizations to personalize interactions, streamline administrative processes, and proactively identify employee needs. According to a Deloitte survey, nearly 80% of executives consider employee experience a key factor in engagement and retention, yet only 22% believe their companies excel in delivering a differentiated and meaningful experience [26]. To address this gap, organizations are increasingly implementing intelligent assistants and AI-powered recommendation systems that provide employees with personalized support, mirroring consumer-grade digital experiences in the workplace.

Recent studies underscore the growing employee demand for intelligent, user-friendly HR technologies. ServiceNow's 2022 annual report found that nearly 30% of employees want HR systems to function with the simplicity and searchability of Google [18]. Similarly, Gartner reported that 47% of HR leaders prioritize improving employee experience, linking it directly to increased performance, organizational loyalty, and lower attrition [27]. Additionally, 16% of employees express interest in voice-activated support tools—such as those similar to Siri—for accessing workplace information. These trends highlight the rising importance of AI-powered solutions, including chatbots and machine learning systems, in delivering personalized, efficient, and responsive employee experiences. Artificial intelligence is significantly transforming the learning industry landscape, particularly within corporate and HR development environments, by enabling personalized and adaptive learning experiences. AI-driven platforms allow organizations to tailor training and upskilling programs based on employee performance, preferences, and learning needs. This shift is part of a broader digital transformation in HRM that prioritizes data-driven, continuous learning strategies [28]. However, studies have shown that despite the growing availability of AI tools, many organizations are not yet prepared to integrate them effectively into strategic HR processes [29]. Challenges such as lack of digital skills, organizational inertia, and ethical concerns often limit implementation and impact [30]. Therefore, the adoption of AI in learning and development must be supported by clear frameworks that align technological capabilities with HR goals, ensuring fairness, transparency, and long-term workforce adaptability.

By monitoring and analyzing data with AI, organizations can better understand employee engagement and concerns, including sensitive areas such as workplace safety and harassment.

Artificial intelligence is increasingly used to improve employee experience and retention through interactive tools such as AI-equipped chatbots and predictive analytics. Chatbots facilitate employee communication by providing instant and personalized responses to frequently asked questions, and have proven effective in onboarding processes, thus reducing administrative workload and increasing engagement [31]. Beyond communication, AI technologies are also being implemented to predict potential resignations. For

instance, some systems monitor digital behavior patterns—such as browsing history or interaction levels—to identify early signs of disengagement [24]. While this predictive capability may help organizations take preventive measures, it also raises ethical concerns regarding employee privacy and autonomy. The use of people analytics tools [32] must be accompanied by transparency and clear data governance policies to avoid mistrust and legal risks. Therefore, while AI offers efficiency and personalization, its integration into HR functions must balance technological advancement with employee rights and organizational values.

In the field of human resource management (HRM), successful employee engagement is a primary objective. Approaches include ensuring workers are actively engaged in their tasks, aligning personal goals with those of the organization and the project, and facilitating quality work while maintaining a healthy work–life balance. Resistance to AI adoption in HRM stems from job security concerns and skepticism about reliability, which organizations can address through clear communication and comprehensive training [33]. Factors such as cultural diversity, organizational policies, the specific context of employees, and the level of diversity, along with compensation policies and employee satisfaction indicators [34], all contribute to creating effective methods for employee engagement. Aligning these elements can support and promote employee involvement. Machine learning techniques can use real-time analytics to monitor issues employees face and identify solutions through sentiment analysis. For example, if a significant number of employees express concerns about delays in travel expense reimbursements, it may indicate operational issues that need resolution.

In the current economic climate, organizational culture is becoming increasingly critical. Managing a successful organizational culture influences team performance, talent attraction, and retention, according to many HRM reports. While some companies adopt formal or informal work cultures, most integrate a combination of both. The importance of organizational culture in understanding the effectiveness of HR processes is particularly significant.

To monitor the dynamics of changes in organizational culture and their impact on employee performance, the use of machine learning models is essential for overall improvement in organizational culture. For instance, developing a machine learning model using training datasets and the nearest neighbor technique can help improve the quality of the system, practices, and organizational culture [35].

Human resource management can leverage machine learning models in a wide variety of ways. Their use can lead to more efficient decision-making within HRM when a company transitions to systems for implementing plans based on more accurate evaluations generated by machine learning models, thereby eliminating the need for human intervention and eliminating human subjectivity [1].

According to Forbes statistics [36], over 40% of employees who leave a job do so within the first year, often departing within the first 90 days. The most common reasons for this decision include a lack of career development opportunities, undesirable aspects of the work environment, and an imbalance between work and personal life. Additionally, many employees experienced burnout due to the pandemic. Approximately 40% of employees reported experiencing burnout during the pandemic, potentially because 37% of them are working longer hours. Burnout appears to be more prevalent among women, with 43% of female leaders affected compared to 31% of male leaders. Also, employee recognition surpasses gaining more autonomy (12%), inspiration (12%), or even a salary increases (7%) as the most essential aspect a workplace can provide to support employee success.

An employee may decide to leave an organization for one or more of the following reasons [2,3,37]:

- A higher salary or a more attractive role offered elsewhere,
- An unsuitable or uncomfortable work environment,
- Misalignment with the company's goals,
- A lack of work–life balance, leading to stress,
- Relocation or pursuit of higher education opportunities,
- A lack of appreciation or feeling underutilized,
- Excessive overtime,
- Observing the departure of talented employees or team leaders from the organization.

Employee retention is a major issue for many companies today. Numerous organizations are exploring various methods to retain their employees, including generating annual reports through surveys to understand their concerns and needs. HR teams aim to identify company-specific turnover reasons and intervene early to retain employees before it is too late, by adopting AI-based applications. Using a machine learning pipeline, several models were tested, including logistic regression, AdaBoost, gradient boosting, and random forest. The most successful algorithm was logistic regression, which provides clear and easy-to-understand explanations for the predictive decisions of machine learning models. This approach enables HR managers to intervene before an employee decides to leave, adjust retention strategies, and take personalized actions [38].

The machine learning algorithm facilitates quick and informed decision-making based on data, achieving a high level of accuracy when trained on data. The implemented model can be retrained and used annually by the HR department through a simple API integrated into the user interface [39].

5. AI-Based Solution Development for Human Resource Management

5.1. Research Findings in HR Explorations

AI-based solutions are increasingly being adopted in human resource management to enhance predictive capabilities in workforce planning, especially in areas such as employee attrition, performance evaluation, and engagement. This section explores how different machine learning models—including random forest, XGBoost, logistic regression, and neural networks—are applied in HR contexts to support evidence-based decision-making. We organize this review according to specific HR challenges, detailing the data sources, modeling techniques, and implications for practice.

Artificial intelligence will significantly influence activities within the human resources (HR) department throughout the entire employee lifecycle. This impact spans HR operations and service delivery, recruitment, learning and development processes, as well as talent management. Initially, the implementation of artificial intelligence will generate new expectations from employees regarding how they interact with HR technologies and the HR department. As this evolves, it will lead to a redefinition of the purpose and structure of roles and teams within the HR department [40].

Employee engagement is key to retaining employees and reducing turnover. Traditional engagement surveys have presented disadvantages such as manual management, limited scope to employee samples, and associated costs. These limitations hinder timely identification and intervention for dissatisfied employees, potentially leading to retention losses.

This is where the power of artificial intelligence comes into play. With current technologies, building a dedicated AI tool can be cost-effective and efficient in anticipating attrition risks, enabling organizations to better predict when employees might leave the company.

The COVID-19 pandemic profoundly altered the workplace, accelerating the adoption of remote work and increasing employee reliance on digital tools. These shifts in working conditions have contributed to a rise in voluntary resignations, widely referred to as "The

Great Resignation" or "The Great Reshuffle," as employees seek greater flexibility and reevaluate career goals. This trend highlights a critical need for human resource departments to adopt data-driven approaches for understanding attrition dynamics. Artificial intelligence (AI) and machine learning models offer the ability to identify the early warning signs of employee dissatisfaction and predict turnover risk with higher accuracy, enabling organizations to take proactive measures to retain talent [41].

These transformations require innovative approaches, such as using multiple classification models to calculate probabilities, employing ensemble models, or using gradient boosting to determine the importance of variables in a model based on the area under curve (AUC) measure. These methods help organizations understand and manage employee needs and expectations in the new professional context. The characteristics analyzed were divided into three categories: personal information, work experience, and attendance rate. The results indicate that among the critical factors negatively influencing employees' decisions to leave a company are overtime hours, workload level, monthly income, and the duration between promotions.

The analysis used the random forest model to identify the main factors influencing attrition and the XGBoost model to classify employee types with a higher likelihood of leaving. Utilizing various machine learning models, eXtreme gradient boosting provided the best results, with an AUC of 0.86. Other models, such as logistic regression and random forest, performed well but slightly less effectively [42]. Additionally, the random forest model proved to be the most accurate in predicting employee attrition, outperforming other models [41].

Numerous extensive studies have been conducted on employee turnover rates, identifying many effective data mining challenges in this field. According to [43], voluntary resignations have accelerated over the past decade. His study reveals that employees tend to change organizations every six years to advance their personal careers. It is crucial for senior management to be alerted in advance; otherwise, the organization may face severe repercussions when top-performing employees decide to leave.

The research utilizes six supervised machine learning algorithms to evaluate a dataset from a revenue cycle management company, containing information on 12,000 employees. The proposed system, called the early warning system (EWS), provides predictions on high-risk employees and graphical visualizations to help companies better manage risks. The XGBoost algorithm delivered the best results, with an AUC of 0.95 and an accuracy of 96%. Other models, such as logistic regression, SVM, and KNN, performed less effectively. The EWS system successfully predicted approximately 45% of departures, and its use helped reduce talent loss and financial risk.

According to the employee turnover statistics from [44], about one-third of new employees resign after just six months of work. This trend causes employers to lose valuable talent and requires continuous efforts for recruitment, training, and regular replacement of staff.

Several studies attempt to predict when and which factors cause employees to leave companies. In calculating the probability of leaving, several characteristics are considered, such as salary, years before promotion, business travel, and environmental conditions. These characteristics can be modified and influence the likelihood of leaving the company. Other characteristics, such as gender or age, are not used because they cannot be changed and do not influence the prediction. Research shows that the probability of leaving a company varies and can be calculated based on different characteristics, providing insight into the most significant factors influencing an employee's decision to stay or leave. Additionally, the probability calculations for employee turnover should rely on multiple cases rather than individual employees who have left the company [25].

The increasing turnover rate highlights one of the most significant challenges in the current workforce. This situation compels employers to focus on the factors driving employees to leave and also on strategies to bring them back.

Shortlister [44] has gathered substantial data on this topic with the specific purpose of highlighting how this large-scale workforce turnover shapes the new generation of workers post-pandemic. Organizations across various industries have implemented technological solutions to counteract employee attrition. A concrete example is an information management company that considers combating attrition as a strategic measure essential for maintaining a high level of customer service, referred to as "best-in-class services". In Ref. [43], the authors implemented a framework called the "Early Warning System" (EWS) with RAG (red, amber, green) indicators; this company significantly reduced employee attrition rates. In the same article, the team led by M. Singh, in collaboration with IBM Watson, developed a framework that identifies the reasons behind employee attrition and detects potential turnover. The model is based on evaluating the cost of attrition, comparing the difference between the expected attrition cost before the retention period (EACB) and the expected attrition cost after the retention period (EACA).

The primary and essential goal of this project is to provide a comprehensive overview, along with the demonstration and evaluation of various machine learning approaches, to detect attrition in a company or public institution. The project aims to offer a robust technical solution and predict the probable reasons behind this phenomenon.

Focusing particularly on voluntary employee turnover, the results obtained can then be used by the human resources department to analyze the main reasons contributing to attrition and subsequently take measures to mitigate them.

The first step in developing applications involves collecting the dataset necessary for implementing and testing algorithms. According to one study, it is beneficial for the HR department to collect data to train algorithms over a three-year period, ideally tracking the career progression of employees. Data cleaning/preprocessing is a crucial step in building a predictive model. This process helps identify and select relevant features influencing the labeling of the classes to be predicted.

The initial critical step in the data cleaning process is understanding the structure of the data. The data provided were either captured using a native system or supplied by the HR team. Most of the data were entered without proper validation, resulting in numerous typographical errors, missing fields, and inconsistent values. Discrepancies are eliminated after converting the text format to UTF-8, standardizing nomenclatures, rectifying ambiguous entries, removing defective data, and filling in missing values.

There are machine learning algorithms that require numerical variables for input and output, while others require non-numerical variables. One-hot encoding is used to transform categorical data into numerical data. This method not only simplifies the calculation process but also allows the model to learn more efficiently.

At the core of the resignation prediction process, which aids HR departments, there are various supervised machine learning algorithms. These algorithms are employed to mitigate negative effects, enabling HR representatives to approach employees with offers that are hard to refuse.

Several studies have been conducted to test the best algorithms using collected datasets. These studies showed that the prediction results between models utilizing linear and non-linear regression are similar. Additionally, it was found that the tree-based regression model offers a slight improvement in prediction compared to linear and non-linear regression models. The support vector machine model demonstrates significantly better results than the previous models, being comparable to those of the random forest model. Furthermore, neural networks achieve the best prediction performance R^2 on the dataset (Table 1) [45].

According to the authors, the ordinary least squares regression and ridge regression models demonstrated equivalent accuracy, both outperforming robust regression in terms of R^2 values.

Table 1. Comparation of model prediction performance.

The Model	The Prediction
Linear regression	
Ordinary least squares regression	0.426
Robust regression	0.423
Ridge regression	0.426
Non-linear regression	
General Gaussian	0.426
General Poisson	0.426
Gamma regression	0.426
Regression tree	0.500
SVM	0.855
Random forest	0.853
Neural network	0.861

The use of the Pearson correlation coefficient is a valuable tool for feature selection in predictive models, particularly when the goal is to simplify the model and reduce redundancy among variables. It provides good performance across various feature selection methods for both balanced and imbalanced datasets [46].

Another study [47] conducted on a dataset distributed by IBM, containing 35 features, tested multiple algorithms:

- Gaussian Naive Bayes;
- Multinomial Naive Bayes;
- Bernoulli Naive Bayes;
- Decision trees;
- Random forest;
- Logistic regression.

The preprocessed data were divided into two subsets: one for training the models and the other for evaluating classifier performance. Evaluation methods included accuracy, precision, recall, and F-score to identify the most effective classifier. The best result was achieved by the Gaussian Naive Bayes algorithm due to its ability to minimize false negatives (with a recall of 70.76%) [47].

5.2. Predictive Approaches

Predictive AI-based research in HR departments enables data-driven decision-making by forecasting employee behavior, enhancing talent management, and optimizing workforce strategies.

Based on data from an HR department dataset of a company [47], available on Kaggle, predictions regarding employee turnover were made. This dataset contains 10 distinct features related to a total of 1470 employees. The information determines whether an employee left or stayed in the company based on these features.

A predictive model was developed using machine learning algorithms, including gradient booster, simple to assess machine learning model, support vector machine, logistic regression, K-nearest neighbors, Gaussian Naïve Bayes, and random forest. The model was trained using 90% of the dataset and tested on the remaining 10%. The most effective

algorithm, random forest, achieved an accuracy of 90.20%, while Naïve Bayes had a lower accuracy of 80.20%.

This study considers a broader range of variables compared to previous methods, focusing on parameters underlying workforce turnover. A detailed analysis of various data analysis methods was conducted to accurately predict employee turnover and the factors contributing to this phenomenon.

Model selection in HRM requires careful consideration of the trade-off between interpretability and predictive performance. Linear models provide clarity but may fall short in complex tasks. Ensemble and deep learning models provide superior accuracy, especially in nonlinear, high-dimensional, or imbalanced datasets, but at the cost of reduced transparency. Therefore, algorithm choice should align with the organization's priorities, regulatory environment, and ability to implement interpretability techniques.

External factors, such as economic conditions, influence business activities, potentially increasing employment rates. Additionally, institutional criteria, such as the field of activity, task completion, organizational characteristics, compensation, hierarchical levels, workplace, evaluation processes, organizational culture, job responsibilities, benefits, and promotions, can also affect turnover rates [16].

It is true that numerous empirical studies have focused on one or at most two aspects influencing performance, such as training, workforce diversity, leadership style, employee engagement, satisfaction, communication, and organizational culture. In contrast to other approaches, another study has focused on adopting the cross-industry standard process for data mining (CRISP-DM) model. This model, published in 1999, aims to standardize, organize, and implement machine learning or data analysis projects across a wide range of industries.

Based on the relevant factors identified in the literature review regarding their influence on employee performance, it was decided to use an HR dataset developed by HR management professors at the New England College of Business. This dataset contains 36 variables describing 311 employee profiles. To evaluate employee performance, a model based on multinomial logistic regression was chosen [16]. Another study was conducted on an HR analytics dataset from IBM, available on Kaggle, to investigate employee attrition history. This synthetic dataset was created by IBM's data science team and contains 1471 records and 34 characteristic variables, grouped into three categories: personal information, work experience, and participation rate. The analyzed characteristics included aspects such as work–life balance and employees' marital status. Using this dataset allowed the exploration and investigation of factors influencing employee attrition in a controlled and synthetic manner, ensuring the confidentiality of the company's internal information. Following the application of multiple algorithms, Xtreme gradient boosting achieved the highest accuracy on the presented data [40].

In another predictive analysis study, machine learning algorithms were applied to HR data available on Kaggle [48]. The dataset includes information about 14,999 employees, based on 10 distinct parameters. Before building an efficient model, the dataset underwent a manual preparation process.

RapidMiner, a comprehensive data science platform characterized by its visual workflow design and full automation, has been successfully used in some studies for HR data analysis [49]. This platform allows the comparison of multiple training models, including support vector machine, Naive Bayes, decision tree, and neural networks. Based on the training data from the dataset and the analyzed parameters, the decision tree model in this study ultimately extracts useful rules through machine learning and model training, which can be used to accurately predict employee turnover rates.

Satisfaction level, salary, and time spent in the company rank highest in influencing employee turnover rates, showing both positive and negative correlations. Satisfaction level, number of projects, average monthly hours, and other attributes have significant effects on employee turnover rates. The analysis process aims to understand the underlying reasons for employee attrition. The predictive model developed in this way can anticipate retention probabilities by considering specific statistical parameters as input for the prediction process [50].

This approach enables the identification of factors contributing to attrition and allows organizations to take preventive measures for retention or intervention in the case of at-risk employees.

The following algorithms were used for performance evaluation: decision tree (DT), support vector machine (SVM), K-nearest neighbor (KNN), Naïve Bayes (NB), and the boosted Ttree ensemble. Interestingly, within this dataset, the decision tree achieved the best performance [50], while SVM (support vector machine) had the poorest performance in predicting employee turnover. This variation in performance may be influenced by the complexity of the data or how well the respective algorithms adapt to the specific characteristics of the dataset. Another study [51] highlighted the best results being achieved with the decision tree algorithm. It is important to analyze these results in depth to understand why certain models perform better than others and how predictions can be improved in the future.

In another dataset analyzed in a separate study [52], after data analysis and cleaning stages, the highest accuracy was achieved by the random forest algorithm.

Another paper [16] presenting a solution on an HR dataset from IBM, based on information from 2020, includes 35 features describing 1467 unique employees. This dataset encompasses a range of HR-relevant information such as age, education level, gender, promotion history, education details, and rating. These features provide a detailed perspective of employees and various aspects of engagement, performance, and retention within the organization. Utilizing this dataset allows for deeper analysis and understanding of the factors influencing employees and their potential to stay with the company.

Adapting the analysis to departmental levels is an excellent approach to better understand the impact of different roles and contexts on employee turnover. When using algorithms such as logistic regression, random forest, and XGBoost to classify turnover based on the IBM dataset, focusing on different cases within departments can offer a more granular and specific understanding of the factors influencing employee attrition rates across various fields or positions.

By making this distinction, it will be possible to observe how specific departmental or role-related variables within the organization can influence employee turnover. This can help identify and more accurately understand retention issues in different contexts and provide useful insights for taking specific measures in each department or role to improve employee retention [25].

Another study [53] adds further complexity by employing data preprocessing techniques, class balancing, and feature selection, offering a more advanced and tailored approach to predictions based on imbalanced datasets using two techniques:

- ADASYN: This creates synthetic instances for the minority class, thereby increasing the proportion of data for employees leaving the company and improving algorithm performance in predictions for this class.
- Undersampling: This reduces the number of observations in the majority class, retaining only a portion to balance the data and achieve high performance in situations with imbalanced datasets.

The use of the bagging method, which trains models on subsets of the dataset to reduce overfitting and improve accuracy, enhances prediction with a higher success rate [54].

Predicting resignations using neural networks could represent a future direction for research in this area. According to one study [55], the use of neural networks and the ensemble neural network model achieved very good accuracy on a large dataset of approximately 500 records.

A relevant real-world implementation of AI in human resource management is presented in [34], where the authors developed a machine learning framework for classifying employee performance levels. Using a dataset provided by faculty members at the New England College of Business—comprising 311 employee records and 36 HR-related features such as job satisfaction, salary level, number of projects, and delay times—the study employed principal component analysis (PCA) for dimensionality reduction followed by a multinomial logistic regression (MLR) model for classification. The resulting system achieved a classification accuracy of 87.3%, effectively categorizing employees into groups such as "low performers," "good performers," and "needs improvement." This case demonstrates how AI models can be operationalized to support HR decision-making, enabling organizations to design targeted interventions for performance management and strategic workforce planning.

A real-world implementation of AI models is presented in [43]. The study is based on data collected from a large information technology (IT) company in India with over 4000 employees. Its main objective is to develop predictive models that identify employees at high risk of voluntary staff turnover. By applying machine learning techniques such as random forest, decision tree, and logistic regression, the authors demonstrate how data-driven approaches can support HR departments in anticipating staff turnover and formulating proactive retention strategies. This study is firmly situated in the context of HR analytics, providing a concrete example of how artificial intelligence can improve strategic workforce planning and decision-making in enterprise environments.

The development of AI-based solutions in this study is grounded in both theoretical and algorithmic foundations. From a theoretical perspective, we integrate three key frameworks from organizational psychology—organizational commitment theory, expectancy theory, and equity theory—to guide model design and interpretation. These theories inform the selection and interpretation of features within the IBM Employee Attrition dataset: affective commitment is reflected in variables such as JobSatisfaction and EnvironmentSatisfaction; expectancy is captured via OverTime and PerformanceRating; and perceptions of equity are modeled through features such as MonthlyIncome and Recognition.

On the algorithmic side, we employ logistic regression, random forest, and XGBoost—each offering complementary advantages in terms of linearity, non-linearity, and interpretability. Feature selection is conducted using Pearson correlation, and class imbalance is addressed through ADASYN and random undersampling. We perform hyperparameter tuning to optimize performance, and evaluate the models using accuracy and AUC.

This combined foundation supports a robust and interdisciplinary framework for employee attrition prediction, enhancing both the theoretical validity and practical relevance of AI-driven HRM solutions.

In this context, AI tools are increasingly being applied to address concrete HR challenges, including employee attrition prediction, performance evaluation, and recruitment optimization. The focus is on real-world applications that support strategic decision-making for HR professionals.

One notable technical challenge in attrition prediction is the frequent class imbalance present in HR datasets. To address this, future research may benefit from integrating hybrid architectures such as SMOTEDNN. Recent work by Joloudari et al. [56] introduced

a framework that combines SMOTE with convolutional neural networks (CNN), achieving 99.08% accuracy across 24 datasets, including structured and tabular formats. Their study highlights the critical role of hyperparameter tuning, both in the oversampling phase (e.g., the number of neighbors in SMOTE) and during DNN training (e.g., learning rate, number of hidden layers, dropout rate). While the current study focused on tree-based and logistic models, incorporating SMOTEDNN-inspired frameworks could significantly improve robustness and accuracy in future HR analytics—especially for identifying rare but impactful attrition cases.

6. Comparative Analysis

6.1. Critical Model Performance Analysis

This subsection provides an in-depth comparative analysis of the machine learning models applied in HRM contexts, focusing on their performance, computational efficiency, and suitability to the structure of HR datasets. It highlights model-specific behaviors, algorithmic optimizations, and key findings from the existing literature that explain observed performance differences.

While Section 4 discussed applications, this section provides a technical comparative analysis of the machine learning models used in HRM. It assesses their relative strengths, weaknesses, and suitability based on accuracy, interpretability, and resource demands. The models introduced in the previous section are now examined comparatively to determine which offer the best performance under different HR contexts.

The studies regarding the identification of employee retention rates, as identified in the research articles analyzed in Section 4, are summarized in the tables below. Those tables include the names of the authors, the technique used, the associated task, and the results obtained. For each identified method, we provide a detailed explanation along with an analysis of their respective advantages and disadvantages. For each identified method, we provide a detailed explanation along with an analysis of their respective advantages and disadvantages.

The studies utilized a diverse range of datasets, both public and proprietary. The paper notes that many of these datasets vary in terms of sample size, class distribution, and variable types. For instance, certain studies [57] analyzed are based on imbalanced datasets, often requiring preprocessing techniques such as SMOTE to enhance learning accuracy. Some datasets incorporate real-world HR records [39], while others are synthetic [41,48] to model specific hypothetical scenarios.

While some papers adopt preprocessing techniques such as SMOTE to mitigate class imbalance—which can reduce indirect bias—there is a general lack of consistent reporting on fairness metrics, bias audits, or mitigation strategies.

The paper also observes that these data sources are characterized by a high dimensionality of features (e.g., job satisfaction, overtime, salary level, department), often with a mixture of categorical and continuous variables, which affects the performance of AI models differently.

Hyperparameter tuning plays a critical role in enhancing the predictive performance of machine learning models, especially in human resource management (HRM) contexts where data can be high-dimensional, unbalanced, or noisy. Unlike parameters learned during training, hyperparameters are configured prior to learning and directly influence how a model generalizes. Appropriate tuning is essential to prevent overfitting or underfitting and to ensure reliable decision-making. In the reviewed study by Ansari et al. [43], the authors emphasize the importance of hyperparameter adjustment in improving classification performance for employee attrition prediction. Specifically, they tune the number of decision trees (n_estimators) in the random forest model and optimize the regularization

parameter (C) in logistic regression. This process allowed them to balance bias-variance trade-offs effectively and achieve more robust predictions. Such examples highlight the necessity of incorporating hyperparameter optimization as a standard practice in HRM-related AI applications.

Beyond neural network architectures, hyperparameter tuning remains a decisive factor in optimizing model performance across various AI techniques. As illustrated in the study [55], tuning key hyperparameters such as the number of hidden layers, neurons per layer, and learning rate was critical to achieving model convergence and avoiding overfitting. The authors adjusted these configurations to align the model's complexity with the available training data, enabling the neural network to effectively capture non-linear relationships pertinent to employee development and retention. This reinforces the broader observation that hyperparameter optimization is not only model-specific but also important in HRM applications that deal with evolving, multi-dimensional employee datasets.

Table 2 leads us to distinguish between several models that have been used in solutions developed in last 5 years, as they are detailed below.

Table 2. Comparative analysis of the research articles.

Studies (Author (Year) [Ref.])	Technique	Task	Results
Andr'e Artelt, et al., 2023 [25]	Random forest, XGBoost	Employee attrition prediction and the generation of counterfactual explanations	An approximate 40% increase in salary would significantly reduce employees' intention to leave. Reducing the time since the last promotion by about 5 years has been associated with a lower probability of departure. A combination of a 20% salary increase and an approximately 50% increase in job satisfaction would decrease the likelihood of employees leaving.
Jia Yuan et al., 2022 [45]	Linear regression, non-linear regression, regression tree, SVM (support vector machine), random forest, neural network	Performance prediction, employee attrition	SVM achieved an AUC of 0.855. Random forest achieved an AUC of 0.853. Neural network achieved an AUC of 0.861. Linear regression achieved an AUC of 0.426. Non-linear regression achieved an AUC of 0.426. Regression tree achieved an AUC of 0.500.
Narayana Darapanen, et al., 2022 [42]	Random forest, XGBoost, support vector machine (SVM), logistic regression and a model ensemble	Employee attrition risk prediction	XGBoost had the best performance with an AUC of 0.86. Ensemble average achieved an AUC of 0.85. Random forest achieved an AUC of 0.81. Logistic regression reached an AUC of 0.83. SVM achieved an AUC of 0.72.
Bobbinpreet Kaurm et al., 2022 [50]	Decision tree (DT), Boosted tree ensemble, K-nearest neighbor (KNN), support vector machine (SVM), Naive Bayes (NB)	employee retention prediction	Decision tree (DT): 98.0% accuracy. Boosted tree ensemble: 97.5% accuracy. K-nearest neighbor (KNN): 93.6% accuracy. Naive Bayes (NB): 90.3% accuracy. Support vector machine (SVM): 78.1% accuracy.
Yong Shi, et al., 2022 [49]	Support vector machine (SVM), Naive Bayes, decision tree, neural network	Employee turnover prediction	Decision tree: accuracy 97.03%, precision 97.54%, recall 98.61%, F1 score 98.07%. Neural network: accuracy 93.88%, precision 95.58%, recall 96.47%, F1 score 96.02% Naive Bayes: accuracy 78.94%, precision 90.43%, recall 80.97%, F1 score 85.44%. Support vector machine (SVM): accuracy 78.11%, precision 79.75%, recall 95.61%, F1 score 86.92%.
Vengai Musanga et al., 2022 [46]	Logistic regression (LR), random forest (RF), gradient boosting machine (GBM), decision tree (DT) and K-nearest neighbors (KNN).	Employee churn prediction	With feature selection (Pearson correlation): RF achieved the highest accuracy, 91.76%. Without feature selection: GBM had the highest accuracy, 87.53%. ROC AUC: RF and GBM showed the best results, with large areas under the ROC curves, indicating a high level of separability.

Table 2. *Cont.*

Studies (Author (Year) [Ref.])	Technique	Task	Results
Heng Zhang, et al., 2018 [54]	Logistic regression, gradient boosted decision tees (GBDT)	Employee turnover prediction	Logistic model accuracy: 87.2% Gradient boosted decision trees accuracy: 89.32%.
Shobhanam Krishna, et al., 2022 [41]	Random forest classifier, AdaBoost classifier	Employee attrition prediction	Random forest classifier accuracy 100%. AdaBoost classifier accuracy 84.95%.
Isha Tewari, et al., 2020 [24]	Random forest (RF), gradient boosting machines (GBM), decision trees (DT), support vector machines (SVM), and k-nearest neighbors (kNN)	Predicting employee attrition	RF achieved an accuracy of 91.76% with feature selection. GBM showed robust performance with an accuracy of 87.53% without feature selection.
Said Achchab, et al., 2021 [21]	Machine learning algorithms: primarily used for predictive analytics in recruitment, retention, and performance assessments. KNN, logistic regression, SVM Natural language processing (NLP): For analyzing text data, such as resumes, cover letters, and employee feedback. Robotic process automation (RPA): for automating repetitive tasks in HR operations, such as data entry, scheduling, and initial candidate screening.	Predicting employee turnover and enhancing HR efficiency	Increased efficiency: AI reduces manual work in HR, leading to time and cost savings. Enhanced decision-making: predictive analytics enables HR departments to identify high-risk turnover employees and make proactive interventions. KNN 95%, logistic regression 95%, SVM 5% Improved recruitment accuracy: AI algorithms, such as NLP-based models, enhance candidate screening and matching processes.
Abdullah Abonamah, et al., 2022 [38]	The system architecture comprises three main components: dashboard service: provides a visual interface to HR professionals, displaying key metrics and predictions. ML API services: uses RESTful APIs for delivering machine learning models, focusing on attrition prediction. Database service: stores HR data using PostgreSQL, integrated via an open database connectivity (ODBC) API to support various DBMS. The ML pipeline uses logistic regression as the main predictive model, and AdaBoost, gradient boosting, support vector machine, linear discriminant analysis	The primary task is predicting employee attrition. This involves: Data preparation: handling class imbalance in the attrition dataset and selecting relevant features based on statistical tests. Model training and calibration: logistic regression was chosen as the best model, calibrated with Platt scaling to interpret predictions as probabilities, especially for cases with high stakes like employee attrition.	The calibrated logistic regression model provided better-balanced predictions by reducing false positives. The accuracy is 0.854—logistic regression, 0.822—AdaBoost, 0.847—gradient boosting, 0.849—support vector machine, 0.849—linear discriminant analysis.
Shweta Pandey, et al., 2022 [52]	Modeling utilizes algorithms like random forest, Naïve Bayes, K-nearest neighbors (KNN), and logistic regression.	Using employee characteristics to predict which employees are suitable for promotion, assisting HR in timely and accurate decision-making.	The random forest model was found to have the highest accuracy for prediction tasks. Random forests—99.6%. KNN—89%. Naïve bayes—73.4%. Logistic regression—60%.
Priyanka Sadana, et al., 2021 [39]	Various models—logistic regression, random forest, decision tree, and gradient boosting—were trained and evaluated to determine the best predictor.	The primary task was to predict employee attrition and identify factors that most influence an employee's decision to stay or leave.	The random forest model outperformed others, achieving high recall (93% post-tuning), indicating reliable predictions for attrition.

Table 2. *Cont.*

Studies (Author (Year) [Ref.])	Technique	Task	Results
Costa, R., et al., 2022 [51]	Machine learning models used are decision tree, AdaBoost, and support vector machine (SVM).	The main task of the model is to predict whether an employee is likely to leave the organization. This allows HR teams to proactively address potential turnover by providing targeted support, incentives, or role adjustments.	The decision tree model achieved the highest accuracy (AUC = 0.78), followed by AdaBoost (AUC = 0.74). SVM was not suitable for this problem, with an AUC of 0.50, indicating it struggled to differentiate between retention and turnover.
Apurva BM HR Analytics, et al., 2020 [43]	Six machine learning models (logistic regression, decision tree, K-nearest neighbors, Naïve Bayes, support vector machine, and XGBoost) were tested. The dataset was split 70:30 for training and testing, and the models were evaluated using metrics like accuracy, F1-score, and ROC-AUC.	Identify employees likely to leave based on factors like happiness index, salary, and work environment.	XGBoost outperformed other models with an accuracy of 96%, AUC of 0.95, and F1-score of 0.96, making it the preferred model for attrition prediction. AUC obtained for models: logistic regression—0.71 decision tree—0.71 K-nearest neighbors—0.56 Naïve Bayes—0. 67 support vector machine—0.53 XGBoost—0.95
Sarah S. Alduayj, et al., 2018 [53]	Support vector machine (SVM), random forest, and K-nearest neighbors (KNN)	Predict employee attrition by analyzing factors that contribute to the likelihood of an employee leaving.	Imbalanced dataset: quadratic SVM performed best with an F1 score of 0.503, but overall, the imbalanced dataset yielded low F1 scores across all models. ADASYN balanced dataset: results significantly improved, with cubic SVM, Gaussian SVM, random forest, and KNN (K = 3) achieving F1 scores between 0.91 and 0.93. This demonstrated the effectiveness of the ADASYN oversampling technique. Feature selection: using feature ranking, top features (like overtime, total working years, and job level) were identified. Random forest achieved an F1 score of 0.909 using the top 12 features, reducing model complexity without sacrificing performance. Undersampled dataset: performance was lower than ADASYN, with Gaussian SVM achieving the best F1 score of 0.738. This indicated that undersampling might lead to information loss, affecting model performance.
Amine Habous, et al., 2021 [47]	Several classification algorithms were used, including: Gaussian Naïve Bayes Bernoulli Naïve Bayes Multinomial Naïve Bayes Decision tree Random forest Logistic regression	The task was to predict employee attrition and minimize false negatives (i.e., accurately identify employees who are at risk of leaving).	The task was to predict employee attrition and minimize false negatives (i.e., accurately identify employees who are at risk of leaving).
Raj Chakraborty, et al., 2021 [16]	algorithms used: random forest, logistic regression, Naïve Bayes, gradient boosting, SVM, and K-nearest neighbors.	Understanding the key factors driving employee turnover and developing efficient predictive models.	The random forest model delivered the best results with an accuracy of 90.2%, while Naïve Bayes had the lowest accuracy at 80.7%.
Preethi Keerthi DSouza, 2023 [58]	Bayesian network	Employee attrition analysis and prediction	Percent correct classification for attrition = 89%

Since this study relies on data and performance metrics reported in the existing literature, the comparative performance of XGBoost—particularly its superior AUC values—can be contextualized through the characteristics of the datasets described in those sources. Notably, several of the reviewed studies [25,42,43] highlight that XGBoost performs especially well in scenarios with imbalanced target distributions and heterogeneous feature relevance. Its gradient boosting mechanism allows it to capture complex, non-linear interactions and mitigate overfitting through regularization, which is particularly effective when working

with high-dimensional HR datasets that include both categorical and continuous variables. These structural advantages explain its enhanced predictive performance, especially when compared to more linear models or ensemble methods based on random bagging.

These structural considerations are further illustrated in Table 3, which summarizes the performance tendencies of selected models based on key dataset characteristics relevant to employee attrition prediction tasks.

Table 3. Comparative characteristics of AI models used in employee attrition prediction.

Model	Relation Type	Sensitivity to Imbalance	AUC
Logistic regression	Linear	High	Lower
Random forest	Non-linear	Medium	Good
XGBoost	Non-linear	Low	Best

As shown in Table 4, XGBoost and random forest consistently demonstrate superior AUC and accuracy values across datasets, which can be attributed to their ensemble architectures and ability to handle feature heterogeneity and imbalance. In contrast, models like logistic regression and SVM perform better under linear and well-scaled conditions but struggle with noisy or high-dimensional inputs.

Table 4. Comparative performance of AI models based on AUC and accuracy.

Model	AUC	Accuracy	Comments
Logistic regression	0.70–0.85	79–82%	Performs well on linear and well-scaled datasets; less effective with complex, nonlinear data.
Random forest	0.80–0.91	78–89%	Stable and interpretable; prone to overfitting without regularization.
Support vector machine (SVM)	0.65–0.88	68–85%	Performance varies significantly across studies.
ExtraTrees	0.83–0.95	80–92%	Improved generalization and robustness to noise; suitable for HR datasets with mixed features.
XGBoost	0.96–0.99	85–94%	High performance in imbalanced and nonlinear contexts; strong regularization reduces overfitting
Neural network	0.84–0.92	81–90%	Good at capturing complex patterns; requires more training time and larger datasets.

As highlighted in this section, the superior performance of XGBoost and random forest in several of the reviewed studies can be attributed to their ensemble-based architecture—an approach in which predictions are generated by combining multiple decision trees, either in parallel (random forest) or sequentially and correctively (XGBoost), to achieve increased accuracy and better generalization capacity. This strategy allows them to efficiently handle high-dimensional, noisy, and imbalanced HR datasets.

Random forest benefits from feature bagging and randomization, which helps reduce variance and prevent overfitting. XGBoost, on the other hand, uses gradient boosting with regularization, allowing it to optimize predictive performance by sequentially learning the tree while controlling model complexity.

Furthermore, both models automatically assign feature importance scores, facilitating interpretability, which is particularly valuable in HRM contexts. These advantages collectively explain their higher AUC and accuracy values compared to linear models such as logistic regression or SVM, especially in datasets with mixed feature types and asymmetric class distributions.

In addition to the superior performance of ensemble-based models, it is also important to address the variability observed in support vector machine (SVM) results across different studies.

Recent studies [13,57] indicate that the performance of support vector machines (SVM) in employee retention prediction varies significantly depending on several interrelated factors, which may explain the inconsistencies observed across published works. Specifically, SVM models are highly sensitive to the structure of the dataset, including the linearity of relationships, feature scaling, and class imbalance. In scenarios where features exhibit complex, non-linear interactions or high dimensionality, models such as XGBoost or ExtraTrees often outperform SVM due to their inherent flexibility and insensitivity to scaling. Additionally, the effectiveness of SVM is closely tied to kernel selection (linear, polynomial, radial basis function), regularization strength (C), and gamma parameters. Improper tuning or default configurations may lead to suboptimal margins, especially in imbalanced datasets or when categorical features are not adequately encoded or scaled. These factors suggest that while SVM can yield strong predictive performance under ideal conditions—such as linearly separable and balanced data—it requires more careful preprocessing and tuning than tree-based ensemble models. Therefore, comparative analysis must consider these contextual dependencies to accurately assess model suitability in HRM applications.

Based on the reviewed studies [59], we observed that XGBoost often achieves higher predictive performance; it also incurs higher computational cost during training due to gradient-boosting operations. However, because XGBoost builds shallower trees and benefits from optimized implementations (including GPU support), it may outperform random forest in prediction speed. These trade-offs suggest that model selection in HRM should consider not only accuracy and interpretability, but also computational efficiency, particularly for large-scale or real-time deployment scenarios.

6.2. Answers to the Research Questions

Based on the critical comparison of the machine learning solutions applied to HRM problems, this subsection provides direct answers to the research questions stated in Section 1. The insights are derived from both our empirical analysis and the reviewed literature.

Studying the applied techniques in this comparative way, the research questions mentioned in the first section will find answers as we conclude below.

Q1. What AI-driven approaches have been implemented in human resource solutions?

AI significantly impacts HR by automating processes, enhancing employee experience, and optimizing strategic decisions. Key benefits and challenges include the following:

- Recruitment and selection: AI algorithms assist in sorting candidates and reducing biases. Machine learning models improve the match between candidates and roles.
- Performance management: AI provides behavior analysis for more accurate evaluations.
- Turnover prediction: Predictive models help HR identify employees at risk of leaving and intervene proactively.

Q2. What models have been designed to enhance database analysis for HR solutions?

As was observed in the previous comparative table, different studies suggest the use of predictive models, each with its distinct characteristics and performances:

- XGBoost and random forest: Proven effective in turnover predictions due to their ability to handle large and complex datasets. XGBoost demonstrated excellent performance with an AUC of 0.86 in general studies and 0.95 in a specific context.
- Logistic regression and SVM: Delivered weaker results in comparison but have the advantage of easy interpretability.
- Neural networks and ensemble models: Achieved the best results for complex predictions, with increased accuracy in studies on large datasets.

In the case of preprocessing and feature selection, the steps of data cleaning and feature selection are critical:

- One-hot encoding and data standardization ensure model compatibility.
- Feature selection based on Pearson correlation reduces redundancy and increases model efficiency.
- Balancing algorithms such as ADASYN and undersampling assist in handling imbalanced datasets.

Q3. How can the efficiency of HR solutions be optimized?

Performance differs according to the complexity of the databases used; a fact that leads to the following findings:

- Ensemble boosting and random forest achieve excellent results on large and complex datasets.
- Linear regression and logistic regression are effective for simpler datasets but are limited in handling more complex predictions.
- Decision trees and SVM show variable results but require advanced preprocessing for high performance.

AI is transforming HR by providing automated and predictive solutions for employee retention. Complex models like XGBoost and neural networks deliver superior results in turnover prediction. Optimal data settings and feature selection contribute to the accuracy of predictive models, making them a strategic tool for HR in talent retention.

Based on the results reviewed, several key patterns emerge in how different machine learning models perform across HR analytics tasks. The comparative analysis of the selected studies highlights notable differences in the performance of machine learning algorithms applied to HR-related tasks. Tree-based ensemble models such as XGBoost and random forest consistently achieved higher accuracy and AUC scores compared to traditional models like logistic regression and SVM. This can be attributed to their capacity to handle non-linear relationships, feature interactions, and imbalanced datasets—common characteristics in HR data, where variables like salary, job role, and promotion frequency often interact in complex ways. On the other hand, linear models tend to underperform in such scenarios due to their assumption of linearity and limited flexibility in feature representation. Furthermore, ensemble methods are generally more robust to noise and missing values, which frequently occur in real-world HR datasets. Neural networks also demonstrated strong predictive capabilities, particularly on larger datasets, but require careful tuning and computational resources. These findings suggest that the choice of algorithms should be guided by the data structure, feature complexity, and interpretability requirements. While high accuracy is desirable, models that provide explainable outputs, such as decision trees or logistic regression, may be more appropriate in contexts where transparency and fairness are critical in HR decision-making.

Although the machine learning techniques employed in this study—such as random forest, XGBoost, and logistic regression—are widely recognized and used in the existing literature, their methodological contribution here lies in the specific way they are adapted and operationalized for the context of employee attrition prediction within human resource management (HRM).

Rather than introducing these models as novel algorithms, we systematically selected and combined preprocessing strategies (e.g., Pearson correlation-based feature selection, ADASYN-based class balancing, and hyperparameter tuning) with these models to enhance both predictive performance and interpretability. Furthermore, we assessed the interaction of these techniques with various model types—linear vs. non-linear, interpretable vs. black-box—and with key dataset characteristics such as high dimensionality and class imbalance.

This integrated methodological approach offers a comparative insight across the models and demonstrates the practical considerations necessary when applying AI techniques in HR settings. By aligning technical model design with the ethical and organizational

constraints of HR, we aim to contribute not only to algorithmic performance evaluation, but also to the broader applicability of AI in data-sensitive domains like workforce analytics.

To complement the comparative performance metrics discussed above, we further investigate the underlying factors contributing to the observed differences between models.

To strengthen the interpretability of the experimental results presented in this section, we provide additional insights into the underlying causes of performance differences across models, considering data structure, feature interactions, and model-specific characteristics. XGBoost achieved the highest AUC values due to its ability to handle high-dimensional, sparse data and model complex nonlinear interactions through additive tree structures. Logistic regression, while simpler, still performed well, particularly for features with near-linear relationships to attrition (e.g., OverTime, JobSatisfaction). In contrast, support vector machines were less effective in this context without kernel optimization, being sensitive to feature scaling and class imbalance.

Furthermore, feature interdependence—quantified using the Pearson correlation coefficient—plays a significant role in influencing model variance and bias. For example, correlated variables such as YearsAtCompany and JobRole reduce the effectiveness of models like Naïve Bayes that assume feature independence. Class imbalance also skews model learning, especially for linear classifiers, and this was mitigated using ADASYN, which improved minority class representation during training.

These findings are supported by the related literature. Chakraborty et al. [16] observed that ensemble methods like random forest outperform simpler models by effectively managing noise and complex interactions. Mourad et al. [34] further emphasized the importance of aligning data preprocessing (e.g., PCA, feature selection) with model architecture to improve classification robustness in high-dimensional HR datasets. These theoretical and empirical perspectives enhance our understanding of why specific models excel or underperform in employee attrition prediction scenarios.

7. Future Perspectives

However, the models reviewed do not incorporate mechanisms for dynamically adjusting career trajectories. A promising direction for future research in AI applications in human resource management (HRM) involves the development of fully automated and adaptive systems for employee career planning. Such systems, powered by generative artificial intelligence, could continuously analyze the evolution of employee competencies, organizational restructuring, and labor market trends to generate personalized and adaptive career paths. Unlike existing tools that focus primarily on enhancing current processes, this approach would allow for real-time recalibration of professional goals, identifying development opportunities aligned with both organizational strategies and individual aspirations.

Moreover, optimization algorithms such as the Levenberg–Marquardt neural network—proven effective in engineering domains—could be explored for their applicability in HRM contexts. These techniques may offer advantages in structured or small-scale HR datasets, where non-gradient-based optimization methods have the potential to enhance prediction accuracy and overall model performance.

Inspired by the model applied in vocational education [60], where generative AI is used to deliver personalized recommendations and adaptive learning paths, such a system could integrate natural language processing for intuitive user interactions and machine learning algorithms to anticipate career developments. By incorporating modules for continuous feedback, skills gap analysis, and career path simulations, the system would support a proactive and resilient HR strategy capable of quickly adapting to new labor market challenges.

To further advance intelligent decision-making in HRM, future research could explore multi-technology integration. Recent studies highlight the added value of combining natural language processing with machine learning models to address more complex HR problems such as sentiment-driven attrition prediction and organizational culture assessment.

Recent research underscores the growing importance of integrating multiple AI technologies to address complex challenges in human resource management (HRM), especially through the combination of natural language processing (NLP) and machine learning (ML) techniques. For instance, the study in [61] investigates whether large language models (LLMs), specifically a fine-tuned GPT-3.5 model, can outperform traditional classifiers such as logistic regression, support vector machines, random forest, and XGBoost in predicting employee attrition. Using the IBM HR Analytics dataset, the authors demonstrate that GPT-3.5 achieves a superior F1-score of 0.92—considerably higher than classical ML models—due to its ability to capture implicit signals from textual data that reflect burnout, dissatisfaction, or disengagement. In a complementary effort, the CultureBERT framework presented in [62] applies transformer-based NLP to classify employee-written text in order to assess organizational culture. By fine-tuning BERT on internal feedback datasets, the model significantly outperforms keyword-matching methods, improving classification accuracy by up to 30 percentage points. Together, these studies reveal that combining LLMs with predictive analytics enhances the ability of AI systems to interpret both structured and unstructured data, offering a more intelligent and context-aware foundation for applications such as turnover prediction, cultural fit assessment, and sentiment analysis in HRM.

Complementing these technical advancements, other recent studies have expanded the scope of AI in human resource management beyond traditional applications such as turnover prediction and recruitment. For example, Murire (2024) [17] demonstrates how AI technologies are increasingly used to shape organizational culture by supporting transparent communication, participatory leadership, and a continuous learning environment. In the domain of career development, Young et al. (2025) [63] utilize HR analytics and natural language processing to recommend personalized development paths aligned with employee competencies and goals. Furthermore, Mullens and Shen (2025) [64] introduce the 2ACT framework, which uses AI to analyze skill gaps and mobility opportunities, generating dynamic internal and external career transition strategies. These approaches signal the growing potential of AI to support long-term employee engagement, organizational agility, and strategic workforce planning.

A promising direction for future research in the field of artificial intelligence applied to human resource management lies in the design of a fully automated and adaptive employee career planning system. Unlike traditional solutions that merely optimize existing career cycle stages, such a system could integrate generative AI and machine learning components to anticipate, model, and adjust career trajectories dynamically, based on both individual and organizational developments.

This approach is supported by a series of recent studies that propose emerging and relevant solutions:

- Antoniuk, Ivens, and Kolyada (2025) [65] examine how artificial intelligence reshapes the HR function through a model of adaptive management, where career development decisions are guided by predictive models and contextual factors. Their study introduces a framework that enables organizations to personalize interventions for each employee by considering performance, feedback, and preferences, thus promoting fluid and individualized career management. The authors highlight the benefits of human–machine strategic collaboration in decision-making processes.

- In another study, Hoa Do, Lin Xiao Chu, and Helen Shipton (2025) propose an AI-driven HRM model aimed at supporting employee resilience. It integrates real-time feedback mechanisms, behavioral monitoring, and adaptive recommendations for training and internal mobility. The model emphasizes AI's capability to detect early signs of burnout, disengagement, or professional stagnation and respond proactively through personalized suggestions. Contextual data and psychosocial indicators are used to continuously adjust retention and development strategies [66].
- Mullens and Shen (2025) introduce the 2ACT (*AI-Accentuated Career Transitions*) framework, which uses AI to build skill bridges between employees' current positions and future career paths. By dynamically analyzing skill gaps and labor market trends, 2ACT generates suggestions for internal or external transitions, supported by personalized learning paths. The model combines natural language processing (NLP) for user interaction with recommendation algorithms to propose concrete steps for career progression [64].

Integrating these ideas into the architecture of a unified AI-guided system could revolutionize the way organizations approach career development. Such a system would enable the continuous recalibration of career goals, simultaneously adapted to employee aspirations and organizational strategy—representing a major step toward a truly data-driven approach to HR.

A critical direction for future research involves the systematic integration of ethical frameworks to ensure that AI applications in human resource management (HRM) remain aligned with principles of transparency, fairness, and accountability. While the current study emphasizes predictive performance and theoretical alignment with organizational psychology, future investigations must address how AI systems can respect employee privacy, support explainability, and ensure responsible decision-making. Recent work by Roth et al. [67] highlights key ethical concerns in AI-enhanced HRM, advocating for the application of GDPR principles, explainable AI methods, and institutional accountability mechanisms to mitigate algorithmic bias and protect stakeholder rights. These considerations will guide real-world HRM case studies, in which we aim to assess how AI-driven recommendations align with the core principles of ethical, responsible, and trustworthy artificial intelligence.

8. Conclusions

This study enhances theoretical understanding by linking AI integration in HRM to established theories like organizational commitment, expectancy, and equity theory. These theories help interpret AI predictions in terms of motivation, engagement, and perceived fairness. Practically, the research guides HR professionals in choosing suitable AI models for tasks like attrition prediction, recruitment, and engagement tracking. A comparative model analysis helps balance complexity, performance, and interpretability to align with strategic HR goals. Machine learning enables early detection of disengagement or potential turnover, allowing for proactive HR interventions. These models also support broader HR functions, such as career development, performance alignment, and employee satisfaction. Predictive analytics improves decision-making and minimizes disruption from unexpected departures. Analyzing employee data using machine learning uncovers key factors influencing turnover and retention trends. This proactive approach aids in improving work conditions and retention strategies. Overall, the study demonstrates the powerful role of AI in transforming HRM through data-driven insights.

To conclude, in this critical review, we conducted a detailed analysis of the integration of artificial intelligence (AI) into human resource management (HRM), with a focus on machine learning (ML) algorithms used for recruitment, employee retention, and improving

organizational performance. The utility of AI in reducing manual effort, enhancing decision-making processes, and optimizing employee–company interaction is explored.

A valuable direction for future research lies in the integration of explainable artificial intelligence (XAI) techniques into HR analytics. While advanced models such as XGBoost and neural networks offer high predictive performance, they often lack transparency. By applying XAI methods such as SHAP or LIME, researchers and practitioners can better understand how specific features influence model predictions, which is crucial in HR contexts where fairness, accountability, and ethical decision-making are essential. Further studies could evaluate the effectiveness of these tools in real-world HR decision-making and explore how explainability impacts trust in and adoption of AI-based systems in organizations.

In addition to the theoretical and comparative contributions of this study, future research could expand on these foundations by designing adaptive, AI-driven career development systems that incorporate generative models, as discussed in the "Future Directions" section. Such innovations promise to enhance the personalization and responsiveness of HRM strategies in dynamic labor market contexts.

In conclusion, while AI offers transformative opportunities for HRM, its success depends on responsible implementation—guided by theory, supported by transparent models, and aligned with ethical values. Future research should focus on improving explainability, addressing algorithmic bias, and validating AI models in diverse organizational contexts.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial intelligence
HRM	Human resource management
HR	Human resource
ML	Machine learning
SVM	Support vector machine
KNN	K-nearest neighbors
AUC	Area under the curve
NLP	Natural language processing
RPA	Robotic process automation
DT	Decision tree
XAI	Explainable artificial intelligence
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
AIHR	Analytics in human resources management
EWS	Early warning system
RAG	Red, amber, green

EACA	Expected attrition cost after the retention period
EACB	Expected attrition cost before the retention period
NB	Naïve Bayes
LR	Logistic regression
RF	Random forest
GBM	Gradient boosting machine
GBDT	Gradient boosted decision trees
ODBC	Open database connectivity
API	Application programming interface
ADASYN	Adaptive synthetic sampling
LIME	Local interpretable model-agnostic explanations
SHAP	Shapley addictive explanations

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